

Joseph Aaron Musara

Software Engineer — Full-Stack & AI Systems

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[Portfolio](#) · [GitHub](#) · [LinkedIn](#)

SUMMARY

Full-stack software engineer who builds AI-powered products end to end — data model, backend APIs, real-time pipelines, frontend, and production deployment. Recent work includes an AI recruitment SaaS (Next.js + Frappe + OpenAI), a real-time voice interview agent on LiveKit, and an offline-first React Native app with on-device TensorFlow Lite inference. Comfortable owning a feature from idea to shipped.

CORE SKILLS

Languages & Frontend: TypeScript, JavaScript, Python, Next.js (App Router), React, React Native (Expo), Tailwind CSS

Backend: Frappe Framework, Django, Node.js, REST API design, authentication, background workers, server actions

AI & Real-Time: OpenAI API, Anthropic API, CV parsing & AI scoring with rationale, NLP workflows, prompt engineering, LiveKit voice agents, Deepgram (STT), Cartesia (TTS), Silero VAD, TensorFlow Lite (on-device CNN)

Data & Storage: PostgreSQL, MariaDB, Firestore, Redis

Infrastructure & DevOps: Docker, Nginx, Supervisor, Linux VPS, Vercel, Firebase, deployment debugging, production monitoring

Other: Geolocation (GPS / Haversine), Leaflet, data visualisation, Paynow integration, usage-based billing design

EXPERIENCE

Software Engineer — Colossal Hub

October 2025 – Present · Zimbabwe (Remote)

- Lead delivery on AI-powered SaaS products end to end: data model, backend APIs, AI workflows, recruiter dashboards, and production deployment.
- Built the AI recruitment platform, the real-time voice interview agent, and the credit-based billing layer with Paynow integration.
- Stack: Next.js, TypeScript, Python (Frappe), OpenAI, LiveKit, Docker, Nginx, Redis.

Full Stack Developer Intern — Vanguard Economics

November 2024 – April 2025 · Rwanda

- Collaborated with IT teams to develop, test, and deploy full-stack applications with predictive and data-driven features for an economics consultancy operating across Rwanda and the wider region.
- Built interactive React front-ends focused on data visualisations for analyst-facing dashboards.
- Developed Django and Node.js back-end services for advanced analytics; optimised data storage and retrieval against larger datasets.

Software Engineer Intern — Spiderbit

July 2024 – November 2024 · Rwanda

- Worked on production software projects as part of the BTech Honours degree requirement.
- Contributed to web systems and backend logic across a professional software team.
- Several of the systems built during this internship are still running in production.

Inhouse Software Developer — Enes Harness Management Services

February 2022 – June 2022 · South Africa

- Designed and shipped a Django stock management system that significantly improved operational performance by alerting on raw-material levels before they could disrupt the production floor.
- Built and maintained internal tooling, the company website, and customer/supplier communication systems.
- Worked directly with non-technical stakeholders, ran user training, and supported the tools post-launch.

SELECTED PROJECTS

AI-Powered Recruitment SaaS Platform — Colossal Hub

2025 – 2026 · Next.js · Frappe · OpenAI · Redis

- Parses uploaded CVs into a consistent candidate schema for search, filtering, and scoring.

- Scores candidates against a job spec with a written rationale **and** a bias-verification pass — recruiters audit the reasoning instead of trusting an opaque number.
- Backend APIs cover jobs, candidates, applications, interviews, and credit-based billing.
- Designed to cut manual screening workload by an estimated **60%+**. Live: [hr-frontend-colossal.vercel.app](#)

Real-Time AI Voice Interview Agent

2026 · [LiveKit](#) · [Deepgram](#) · [OpenAI](#) · [Cartesia](#) · [Silero VAD](#)

- Designed the full real-time pipeline: room provisioning, Python agent runtime, Deepgram streaming STT, OpenAI question planning, Cartesia TTS, Silero VAD for natural turn-taking, LiveKit Egress for recording.
- Tuned end-to-end response **under ~600 ms** through streaming and overlapping TTS with the next-question plan.
- Session lifecycle designed so dropped agents and network blips don't leave half-open records.

AgriSphere — Smart Agriculture Mobile Platform

2026 · [React Native \(Expo\)](#) · [TensorFlow Lite](#) · [Firestore](#)

- On-device TensorFlow Lite CNN classifies leaf disease (Healthy / Early Blight / Late Blight) with confidence and management advice — targets a **3–5 s inference budget** on average Android hardware.
- Stage-based advisories driven by potato variety (BP1, Diamond, Mont Claire) and planting date.
- Offline-first: capture, classify, advise, and queue writes with no network; Firestore syncs automatically on reconnect. Localisation-ready (English, Shona, Ndebele).

Community Health Reporting & Analytics Platform

2024 — 2026 · [Django](#) · [Leaflet](#) · [NLP](#) · [Mobile-First](#)

- Public-health platform for anonymous community incident reporting, health-worker verification, and hotspot analytics.
- Leaflet-based hotspot map; NLP on free-text fields surfaces clusters and likely duplicates.
- Built mobile-first for poor connectivity in Zimbabwean reporting contexts.

Usage-Based Billing & Paynow Integration

2026 · [Credit-based billing](#) · [Paynow](#) · [AI metering](#)

- Paynow payment initiation, status polling, and transaction lifecycle tracking.
- Per-feature usage logging so AI calls are metered against a workspace credit balance **before** they go out — keeping unit economics visible from day one.

EDUCATION

BTech Honours, Software Engineering — Harare Institute of Technology (*Expected 2026*)